Radio Check

Veteran Support Platform

Project Overview & Funding Proposal

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Safety Testing: Zentrafuge Limited

Platform: Web, iOS, Android

> "The thing that's there at 2am when no one else is."

Executive Summary

Radio Check is a veteran support platform that combines AI companions — who talk like real people, not therapists, not chatbots — with human counsellors, peer supporters, and a multi-layered safeguarding system independently tested across six rounds by Zentrafuge Limited, a specialist safety testing company.

It covers mental health, housing, money, legal, addiction, family, faith, fitness, women's issues, LGBTQ+ support, Commonwealth veterans, service children, serious illness, criminal justice, and crisis intervention — all from a single app that a veteran can access without registering, without giving their name, and without anyone knowing they're using it.

For many veterans, the barrier isn't that help doesn't exist — it's that asking for it feels impossible. Radio Check removes that barrier. There's no waiting list, no referral form, no receptionist, no "your call is important to us." A veteran can open the app at 2am, talk to someone who sounds like them, and if things get serious, a real human is already being notified — with the veteran's location, their conversation context, and the ability to call them directly from the staff portal.

> **The governing test for every AI interaction:** "Would a mate say this on a sofa at 2am?" If no — rewrite.

21	63	6	126k+
AI PERSONAS	APP SCREENS	SAFETY TEST ROUNDS	LINES OF CODE

The Human Side

Counsellors

Qualified counsellors are integrated directly into the platform — not a referral link to an external service. When a veteran is in crisis, or when the AI recognises it is out of its depth, it offers the counsellor option alongside peer support and national helplines. Counsellors have their own portal with:

- **Live chat** with veterans in real time
- **Callback management** — veterans request a call and counsellors pick up the request

- **Case management** — track ongoing cases, add notes, log supervision
- **Safeguarding alerts** — real-time notifications when a crisis is detected, including the veteran's location
- **Panic button alerts** — immediate notification with GPS coordinates
- **Shift rota** — managed scheduling so coverage is always clear
- **Team view** — see who's online and available

Peer Support Network

Real veterans supporting real veterans. The platform allows people to register as peer supporters and go through a structured training programme (see Training section). Peer supporters have their own portal with dashboards, callbacks, alerts, and team coordination. The distinction between peer supporter and counsellor is maintained throughout — different training, different scope, different portal views.

Virtual Coffee Mornings & Community Events

The platform includes a full events system with Agora video integration. Staff can schedule virtual coffee mornings, group sessions, and community events with recurring schedules, participant limits, waiting rooms, and moderation controls.

This is designed for the veterans who are isolated — the ones who won't walk into a drop-in centre but might join a video call from their sofa.

Locating People in Crisis

When a veteran triggers a crisis response (panic button, or the AI detects imminent risk), the platform attempts to locate them through two methods:

> **1. IP Geolocation** — Automatic, no permission needed. Uses the veteran's IP address to determine approximate city/region. This happens silently on every safeguarding alert.
>
> **2. GPS Location** — If the veteran grants browser location permission during the crisis flow, precise GPS coordinates (latitude, longitude, accuracy radius) are captured and attached to the safeguarding alert, overwriting the less accurate IP-based location.

The staff portal includes a **live location map** (Leaflet) showing active users with location data, recent alert locations, regional heatmaps across UK regions, and city/country breakdowns.

When a counsellor receives a panic alert, they can see where the veteran is — not just their name and message, but their physical location on a map. Combined with simultaneous email notifications to all counsellors and admins, this creates a response chain that moves from **detection to location to human intervention in seconds**.

The Training System (LMS)

The platform includes a full Learning Management System for training peer supporters and staff. It is not a placeholder — it is a structured course comprising:

- **Multiple modules** covering Mental Health Awareness, Ethics & Boundaries, Crisis Support & Suicide Awareness, Safeguarding, Communication Skills, and more
- **Mr Clark** — an AI course tutor (GPT-4o powered) who evaluates written reflective responses using a competency-based framework. He assesses against specific competencies and provides formative feedback
- **Reflection questions** on critical modules — learners must demonstrate understanding, not just tick boxes
- **Certificate generation** upon course completion
- **LMS Admin portal** — manage modules, review learner progress, approve completions
- **Automatic flagging** — if a learner's response raises concerns about their readiness, it is flagged for human review

The protocols that govern the AI (spine, grief, identity) are the same principles taught to the humans.

The 21 AI Personas

Every persona inherits the Soul Document (480 lines of behavioural rules) and the modular protocol system. Each has a unique voice, background, and specialist knowledge. They are not generic chatbots wearing different skins — they are distinct characters with distinct purposes.

PERSONA	ROLE	SPECIALISM
Tommy	Lead Persona	General peer support — the most heavily tested persona across all 6 rounds
Grace	Welcome & Signposting	Front door persona — helps veterans find the right support
Bob	Ex-Para Peer Support	Infantry/airborne veteran, straight-talking peer support
Frankie	PTI — Physical Training	Fitness, routine, structure — the one who gets you moving
Margie	Addiction Support	Substance misuse, recovery, harm reduction
Megan	Women Veterans	MST, gender-specific issues, Salute Her UK
Rachel	Criminal Justice	RMP background, courts, probation, veterans courts
Finch	Legal Specialist	Military law, service complaints, legal referrals
Penny	Benefits & Money	PIP, Universal Credit, war pensions, debt
Jack	Compensation	AFCS, War Pension Scheme claims
Rita	Family Support	Families, relationships, impact of service on loved ones
Sam	Forces Kids Specialist	Service children, young carers, youth support
Helen	Carer Support	Caring for veterans, respite, carer's allowance
Alex	LGBTQ+ Veterans	Ban era, identity, community, Fighting With Pride
Kofi	Commonwealth Veterans	Immigration, settlement, right to remain
James	Faith & Spiritual	Chaplaincy, moral injury, spiritual support
Catherine	Thoughtful Support	Calm, intelligent, reflective conversations
Dave	Men's Health	Physical health, prostate, heart, "blokes don't talk" culture

Baz	Housing & Transition	Homelessness, housing applications, transition from service
Mo	Recovery Support	PTSD recovery, rehabilitation, adjustment
Reg	Serious Illness	Terminal illness, end of life, legacy, dignity

What's On Offer — Every Page

Core Features

FEATURE	DESCRIPTION
AI Buddies	Gallery of all 21 AI personas — choose who to talk to based on what you need
Crisis Support	Immediate crisis resources, panic button, Samaritans / Combat Stress / 999
Callback	Request a callback from a counsellor or peer supporter
Live Chat	Real-time chat with human staff

Mental Health & Wellbeing

FEATURE	DESCRIPTION
Mental Health Screening	PHQ-9 depression screening questionnaire with scoring and guidance
Breathing Game	Guided breathing exercises (box breathing, 4-7-8) — gamified with unlockable patterns
Grounding	Grounding techniques for anxiety and panic
Self-Care	Self-care resources and strategies
Journal	Private journal with mood tagging — stored locally, never sent to a server
Mood Tracking	Track and monitor mood over time
Recovery Support	Resources for those in recovery
Substance Support	Addiction and substance misuse resources

Specialist Support Areas

FEATURE	DESCRIPTION
Money & Benefits	Benefits guidance, PIP, Universal Credit, war pensions
Compensation Schemes	AFCS, War Pension Scheme guidance
Criminal Justice	Support for veterans in the criminal justice system

Serious Illness	Support for veterans with serious or terminal illness
Women Veterans	Issues specific to women who served
Commonwealth Veterans	Immigration, visa, settlement rights
LGBTQ+ Support	Support for LGBTQ+ veterans including ban era
Forces Kids	Support for children of service personnel
Family & Friends	Support for families, including raising concerns about a veteran
For Carers	Resources for those caring for veterans
Faith & Service	Spiritual and faith-based support
Historical Investigations	Support for veterans involved in legacy investigations

Community & Resources

FEATURE	DESCRIPTION
Organisations	Directory of veteran support organisations
Regimental Associations	Find your regimental association
Local Services	Location-based local support services
Recommended Reads	Books and reading resources
Podcasts	Veteran-related podcasts
Gym	Fitness resources and gym partnerships

The Safeguarding System

This is not a keyword filter. It is a **five-layer safety architecture** comprising nearly 5,000 lines of dedicated safety code, independently tested across six rounds by Zentrafuge Limited.

The Five Layers

LAYER	FUNCTION	LINES
Text Normaliser	Cleans slang, abbreviations, and obfuscated text so the safety system reads what the veteran actually meant	339
Phrase Dataset	530 phrases across 14 categories, weighted by severity — distinguishes 40+ gradations of distress	780
Safety Monitor	Keyword and phrase matching with negation detection — "I'm NOT going to hurt myself" is correctly identified as a denial	700
Semantic Model	Goes beyond keywords to understand meaning — catches indirect disclosures without obvious crisis words	598
Conversation Monitor	Tracks trajectory across an entire conversation — a conversation that gradually darkens triggers escalation even if no single message crosses the threshold	1,014

A **Unified Safety System** (565 lines) orchestrates all layers and decides: GREEN (normal), AMBER (tone shift, threshold 45), or RED (full crisis overlay with Samaritans, Combat Stress, 999, and platform counsellors, threshold 120).

The Modular Protocol System

Seven protocol files inject real-time behavioural rules based on what the veteran is actually saying:

- **hard_stop.md** — Always loaded. Banned phrases, absolute rules
- **grief.md** — Stay with the dead person, don't pivot to the user's feelings
- **spine.md** — Two-pushback minimum before clean exit; brush-off detection
- **identity.md** — Handle "are you real?" with honesty, not welfare framing; privacy exemption for data questions
- **attachment.md** — Handle romantic/dependency signals with dignity and honesty
- **venting.md** — Hold space for anger, don't interrogate
- **darkhumour.md** — Match gallows humour, don't pathologise it

Identity Dampening & Crisis Override

The system tracks when a veteran is having a philosophical conversation about AI identity. During these conversations, safety escalation is dampened — because scepticism is not distress. But if genuine crisis language appears mid-conversation, a hard override fires immediately, clearing the dampening and activating full crisis response.

What Happens in a Crisis — From AI Score to Human Response

Every message a veteran sends to any of the 21 AI personas is scored in real time by the five-layer safety system. The score accumulates across the conversation — a single dark joke scores low, but a conversation that gradually reveals hopelessness, isolation, and intent will cross the threshold even if no single message contains an obvious crisis keyword.

> **1.** Veteran sends a message → all five safety layers score it simultaneously
>
> **2.** If cumulative score crosses AMBER (45): AI shifts tone, becomes more present and grounded
>
> **3.** If cumulative score crosses RED (120): full crisis overlay appears with Samaritans, Combat Stress, 999, and platform counsellors
>
> **4.** Safeguarding alert is automatically created in the database with IP geolocation
>
> **5.** If veteran grants browser permission → precise GPS coordinates are captured
>
> **6.** Email notification sent instantly to all counsellors and admins via Resend
>
> **7.** Alert appears in the staff portal with the veteran's location on a live map
>
> **8.** Staff can initiate a phone call directly from the portal via Twilio — browser-to-phone, no personal numbers exposed
>
> **9.** Staff can also open a live chat session with the veteran in real time
>
> **10.** The alert is tracked through to resolution — acknowledged, assigned, and closed

Twilio Voice Integration

The staff portal includes **browser-to-phone calling via Twilio**. When a safeguarding alert comes in, counsellors don't need to pick up a personal phone — they can call the veteran directly from their browser. The call is made through Twilio's infrastructure with WebRTC, meaning:

- No personal phone numbers are exposed on either side
- Calls can be initiated from any device with a browser
- Call logs are recorded in the platform for audit and governance
- Staff can also receive inbound calls through the same system

This means the response chain from AI detection to a human voice on the phone can happen in **minutes, not hours**. The veteran doesn't need to call anyone — someone calls them.

Testing Rigour

Six rounds of independent testing by Zentrafuge Limited. Each round comprises structured scenarios designed to break the system — grief conversations, brush-offs, identity challenges, and dark humour. Findings feed directly into protocol fixes. Safeguarding-critical files carry "DO NOT MODIFY" headers and require Zentrafuge sign-off. A CODEOWNERS file enforces review on GitHub, and a CI pipeline runs the safety test suite on every push.

> **Modular Architecture Test Results: 39/41 PASS (95%)**
>
> 1 accepted false positive — safety system over-accumulation on a sustained identity challenge, accepted as safer than a false negative.

The Admin & Staff Portal

A full Next.js administration system with 34 admin tabs and 10 staff tabs covering every aspect of platform operation:

Admin Capabilities

TAB	FUNCTION
AI Personas	Manage and configure all 21 AI personas
AI Usage	Monitor conversation volumes, costs, and patterns
CMS	Content management for app pages
Events	Schedule and manage virtual coffee mornings and community events
Governance	Approvals, compliance, hazards, incidents, KPIs, moderation, reports
Learning (LMS)	Module management, feedback, patterns, approval queue
Logs	Audit trails, callbacks, calls, chats, panic alerts, safeguarding, screening
Rota	Staff shift scheduling
Staff Management	Add, edit, manage staff accounts and roles
Settings	Site configuration, notification emails, branding
Time Tracking	Staff hours and availability
Beta Testing	Manage beta test groups and feedback
Compliance	Regulatory compliance tracking

Staff Portal Capabilities

TAB	FUNCTION
Dashboard	Overview of active alerts, pending callbacks, and team status
Alerts	Real-time safeguarding and panic alerts with location data
Callbacks	Manage veteran callback requests
Cases	Ongoing case management with notes

Live Chat	Real-time chat with veterans
Events	Manage and join community events
Notes	Case notes and documentation
Rota	View and manage shift schedule
Supervision	Clinical supervision logging
Team	Team member status and coordination

Development Cost & Timeline

What Was Built

67,509	31,208	22,525	4,977
BACKEND (PYTHON)	FRONTEND (REACT NATIVE)	ADMIN PORTAL (NEXT.JS)	SAFETY SYSTEM

Total: 126,000+ lines of production code across three applications (veteran app, staff portal, admin portal), plus 21 AI persona definitions, 7 behavioural protocol files, a 480-line Soul Document, and a full LMS with AI-powered assessment.

Traditional Development Cost

A traditional team to build this would typically comprise:

- 2 backend developers
- 2 frontend/mobile developers
- 1 AI/ML engineer
- 1 QA/safety tester
- 1 project manager
- 1 UX designer

At UK agency rates (£500–800/day per developer), a team of 8 working for 6–9 months:

> **Estimated traditional build cost: £500,000 – £900,000**
>
> At freelance rates: £250,000 – £400,000 (with coordination overhead adding time)

How Radio Check Was Actually Built

> **Built by one person**, with AI-assisted development (Emergent platform) and independent safety testing by Zentrafuge Limited.
>
> **Actual development time: approximately 8–10 weeks** of active development.
>
> Achieved at a fraction of the traditional cost.

That is not just a cost saving — it is the difference between "we'll put together a funding proposal" and "it's already built."

Monthly Running Costs

SERVICE	COST	NOTES
AI (GPT-4o)	Variable	Pay-per-use, primary variable cost
Email (Resend)	Minimal	Pay-per-email, safeguarding alerts and notifications
Voice Calls (Twilio)	Variable	Pay-per-minute, browser-to-phone calling for staff
Video (Agora)	Variable	Pay-per-minute for community events
Hosting	Fixed	Standard cloud hosting
Database (MongoDB)	Fixed	Single instance

Estimated total at moderate usage (hundreds of veterans, daily conversations): £50–150/month.

No per-seat licenses. No enterprise subscriptions. No vendor lock-in. The entire platform runs on open-source frameworks with pay-as-you-go services.

What This Means

A veteran who is struggling at 2am has 21 people to talk to who understand service, who won't judge, who won't fold at the first pushback, who know the difference between a joke and a cry for help, and who can connect them to real humans — counsellors, peers, and emergency services — the moment it matters.

The families have a way in. The peer supporters have training. The counsellors have tools. And the whole thing is wrapped in a safeguarding system that has been deliberately, repeatedly, and independently broken and fixed until it holds.

That's Radio Check.